INCH-POUND

MIL-C-82922(OS)
15 November 1994

MILITARY SPECIFICATION

CARTRIDGE, IMPULSE, CCU-123/E (EARTH ANCHOR CARTRIDGE)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the manufacture and acceptance of the CCU-123/E Impulse Cartridge, referred to herein as "cartridge." This cartridge is also known as the Earth Anchor Cartridge, and is used in the LEA-20 Lightweight Earth Anchor.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Shipping, Fiberboard

MILITARY

MIL-P-116	Preservation, Methods of
MIL-C-440	Compositions A-3 and A-4

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by sending a letter.

AMSC N/A FSC 1377

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MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices
MIL-C-45468	Cap, Blasting, Electric: M6 Parts for, and Loading, Assembling and Packing

STANDARDS

MILITARY

MIL-STD-129	Marking for Shipment and Storage - Ammunition and Explosives
MIL-STD-453	Inspection, Radiographic
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-1168	Ammunition Lot Numbering

(Unless otherwise indicated, copies of Federal and military specifications and standards are available from: Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

NAVAL AIR SYSTEMS COMMAND (CAGE Code 30003)

DL 837AS247	Cartridge, Impulse, CCU-123/E
837AS252	Charge
837AS256	Retainer

NAVAL AIR ENGINEERING CENTER (CAGE Code 80020)

615093	Earth Anchor Assembly, Lightweight

(Application for copies of drawings should be addressed to Commander, Indian Head Division, Naval Surface Warfare Center, Data Control Branch (Code 8410), Indian Head Division, Indian Head, MD 20640-5035.)

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CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to the first article inspection (see 4.2.1 and 6.3).

3.2 Primary components. The charge is the primary component. Only one lot of the primary component shall be used in a cartridge lot. One primary lot may be used in more than one cartridge lot. Copies of all material certifications for the primary component shall be supplied to the contracting activity. These certifications shall include, but not be limited to, explosive composition, compliance with MIL-C-440, and pressed density. All requirements of Drawing 837AS252 apply.

3.3 Cartridge production. The cartridges shall be manufactured in accordance with DL 837AS247 and the documents listed thereon. Each cartridge shall be permanently marked with a serial number in the location specified by Drawing 837AS247. The serial numbers of each lot shall be in consecutive order beginning with the number 001. Each cartridge shall meet the requirements of 3.3.1 and 3.3.2.

3.3.1 Visual inspection. Cartridges shall be free of the following defects: illegible, missing or inaccurate identification markings, burrs, dents, sharp edges, or other defects which may prevent entry of the cartridge into the Lightweight Earth Anchor Assembly (Drawing 615093) or affect the resistance of the cartridge to moisture. Each cartridge shall meet the requirements of 3.2, 3.5 and Drawing 837AS247.

3.3.2 Radiographic examination. When radiographically examined in accordance with 4.3.2, the cartridge shall show presence and proper assembly of charge, metal and non-metal parts, proper crimping, sealing, and no foreign materials.

3.4 Cartridge performance. Cartridges shall be capable of meeting all requirements of 3.4.1 through 3.4.4.

3.4.1 Transportation vibration. The cartridge shall not fire or incur damage, degradation or permanent deformation when a transportation vibration cycle is applied in accordance with 4.3.4.

3.4.2 Temperature and humidity cycling. The cartridge shall not fire or incur damage, degradation or permanent deformation when a temperature and humidity cycle is applied in accordance with 4.3.5.

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3.4.3 Transit shock. The cartridge shall not fire or incur internal or major external damage, degradation or permanent deformation when a transit shock test is applied in accordance with 4.3.6.

3.4.4 Functional. The cartridge shall be detonated by an M6 Electric Blasting Cap (MIL-C-45468) when tested in accordance with 4.3.7.

3.5 Workmanship. The cartridge shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be directed to dimensions, finishes, sealing, and assembly operations.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order (see 6.2), the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contractor. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1)
b. Quality conformance inspection (see 4.2.2)

4.2.1 First article inspection. Unless otherwise specified in the contract or purchase order (see 6.2), a first article sample of 15 cartridges shall be subjected to the first article tests of table I. Twelve cartridges shall be expended in the tests listed in table I and three cartridges shall be retained for investigative purposes. Any damage inflicted by the environmental treatments, which would adversely affect the performance of the item in service application shall be cause for rejection of the first article sample. Any further production prior to notification by the contracting agency of first article sample acceptability shall be at the contractor's risk. Failure of any cartridge to comply with the requirements of section 3 shall be cause for rejection of the first article represented.

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TABLE I. First article inspections.

Test Sequence	Test Paragraph	Requirement Paragraph	Sample Group			
			I	II	III	IV
1. Visual	4.3.1	3.3.1, 3.5 & 5	3	3	3	3
2. Radiographic	4.3.2	3.3.2	3	3	3	3
3. Primary component certification	4.3.3	3.2	3	3	3	3
4. Transportation Vibration	4.3.4	3.4.1	3			
5. Temperature and Humidity Cycling[1/]	4.3.5	3.4.2		3		
6. Transit Shock[1/]	4.3.6	3.4.3			3	
7. Functional Test: Ambient Temp.	4.3.7	3.4.4	3	3	3	3

[1/] A visual and x-ray inspection shall be conducted following the specified test.

4.2.2 Quality conformance inspection. All cartridges manufactured under the contract shall be inspected as specified in table II. Cartridges failing to meet the requirements listed in table II shall be rejected and removed from the lot.

TABLE II. Quality conformance inspections.

Test Sequence	Test Paragraph	Requirement Paragraph
1. Visual	4.3.1	3.3.1, 3.5, and Section 5
2. Radiographic	4.3.2	3.3.2
3. Primary component certification	4.3.3	3.2

4.3 Inspections and tests.

4.3.1 Visual inspections.

4.3.1.1 Cartridge inspection. The external condition and appearance of the cartridges shall be determined by comparison to Drawing 837AS247. Each cartridge shall meet the requirements of 3.3.1 and 3.5. Cartridges failing to meet the requirements of 3.3.1 and 3.5 shall be rejected and removed from the lot.

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4.3.1.2 Packaging inspection. The condition of the packaging (inner and intermediate containers), packing (outer container), and markings shall be determined by comparison to the requirements of section 5. Packaging, packing, and markings failing to meet the requirements of section 5 shall be rejected.

4.3.2 Radiographic examination. Radiographic examination shall include x-ray in accordance with MIL-STD-453. The cartridges shall be positioned for the most revealing exposure with the long axis perpendicular to the radiographic beam. Each radiographic plate shall contain one identified inert cartridge to be used as a comparison aid. The cartridges shall be arranged on boards or trays in consecutive order by serial number. Each serial number and any missing serial numbers shall be identified on the radiographic plate. Each radiograph shall carry a permanent identification of the items displayed thereon in a 4-inch by 6-inch region maximum. The radiographic identification shall include the drawing number, the complete lot number in accordance with MIL-STD-1168, the contract number, and the span of serial numbers displayed. Radiographs of the entire production lot shall accompany the ballistic sample to the activity conducting the tests. Any observable imperfections as outlined in 3.3.2, or any other identifiable defects, shall be cause for rejection of the cartridge. Defective cartridges found during radiographic review are to be marked on the radiographic plate and removed from the production lot.

4.3.3 Primary component certification. The charge shall be manufactured of material complying with the specifications indicated in 3.2. Material certification to these specifications is required. The pressed charge shall be dimensionally inspected and weighed to calculate pressed density of the material. The density shall be provided as part of the primary component certification.

4.3.4 Transportation vibration test. The cartridge shall be subjected to vibration testing in accordance with MIL-STD-810, Method 514.4, Category 1 - Basic Transportation Vibration. The test program shall consist of two parts, described in 4.3.4.1 and 4.3.4.2. Each part shall be random vibration for secured cargo, but will vary for the type of carrier, composite wheeled vehicle and a composite two-wheeled trailer. Failure of any cartridge to comply with the requirements of 3.4.1 shall be cause for rejection of the sample or lot represented.

4.3.4.1 Part I - Composite two wheeled trailer. Part I shall be conducted at test levels indicated in Table 514.4-AI of MIL-STD-810, with the spectral shape in Figure 514.4-4 of MIL-STD-810. Every axis shall be tested at each of the following temperatures, −60, 70, and 160°F. The test duration per axis/temperature shall be 96 minutes, which is equivalent to 32 miles.

4.3.4.2 Part II - Composite wheeled trailer. Part II shall be conducted at test levels indicated in Table 514.4-AII of MIL-STD-810, with the spectral shape in Figure 514.4-4 of MIL-STD-810. Every axis shall be tested at each of the following temperatures, −60, 70, and 160°F. The test duration per axis/temperature shall be 120 minutes, which is equivalent to 500 miles.

4.3.5 Temperature and humidity cycling. The temperature and humidity cycling test shall be conducted in accordance with the temperature and humidity cycling test of MIL-D-21625F except that no intermediate withdrawals shall be made. Failure of any cartridge to comply with the requirements of 3.4.2 shall be cause for rejection of the sample or lot represented.

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4.3.6 Transit shock test. The transit shock test shall be conducted in accordance with Method 516.4, Procedure IV of MIL-STD-810, with the following exceptions. The cartridges shall be dropped from a height of four feet onto a two inch thick steel plate supported by reinforced concrete. Each cartridge shall be dropped a total of five times, once on each of the five impact points. All impact points are given in Figure 1. Failure of any cartridge to comply with the requirements of 3.4.3 shall be cause for rejection of the sample or lot represented.

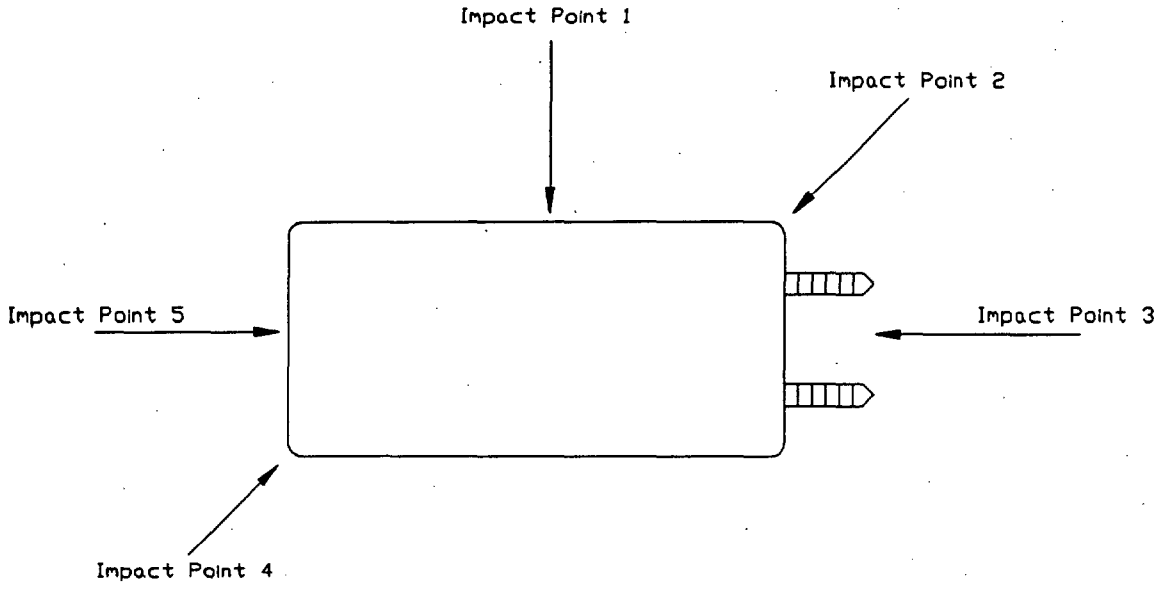


FIGURE 1. Transit shock impact orientation.

4.3.7 Functional test. Sample cartridges for the functional test shall be fired at ambient temperature for the day of the testing. The testing shall demonstrate that the cartridge provides a high- order detonation when initiated by an M6 Electric Blasting Cap. Prior to initiation, the blasting cap shall be inserted into the tube, down the center of the cartridge. A retainer, part number 837AS252, shall be placed over the retainer pins in the cartridge to ensure the blasting cap cannot be inadvertently removed. A current of 1 ampere minimum shall be applied to the M6 Electric Blasting Cap for initiation. Failure of any cartridge to comply with the requirements of 3.4.4 shall be cause for rejection of the sample or lot represented.

4.3.7.1 Test failure. If a test failure is attributable to an assignable cause, excluding the test cartridges, the original test results shall be discarded and the test reconducted.

5. PACKAGING

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be Level A and shall be in accordance with 49 CFR 100-199. The contractor shall also comply with all applicable transportation regulations.

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5.1.1 Level A packaging. Level A preservation and packing (see 6.6), as specified in 5.1.1.1 and 5.1.1.2, shall be used for packaging of all production lot for service use. Unless otherwise specified in the contract or purchase order (see 6.2), the number of inner containers per pack is one.

5.1.1.1 Inner container. Ten cartridges shall be placed in a single polystyrene block, 10-1/4" length, 4-3/4" width, 4-3/4" depth. The polystyrene block shall have ten holes, 1-5/8" diameter each, to accept the cartridges. Also, the block shall have a 2-1/4" by 1/2" by 1/2" deep slot located in the center of the top to accept ten retainers (P/N 30003-837AS256). This block, with the cartridges and retainers, shall have a 0.35" thick sheet of polystyrene placed below and above it, and all of this shall be placed into a fiberboard shipping box conforming to PPP-B-636, Class WR, Style RSC, Type CF, Variety SW, Grade V3c, and internal dimensions 10-1/4" length by 4-3/4" width by 4-3/4" depth. A tight pack is mandatory for proper protection. The fiberboard box shall be closed per Method IV of PPP-B-636, then sealed in a bag for preservation in accordance with Submethod IA-14 of MIL-P-116.

5.1.1.2 Outer pack. Cartridges packaged in accordance with 5.1.1.1 shall be packaged in a nailed wood box conforming to PPP-B-621, Class 1, Style 1, Grade A, with internal dimensions of 12-1/2" length by 7" width by 7" depth. One-inch-thick polyurethane foam sheets shall be placed all around the inner container when it is placed into the outer pack. A tight pack is mandatory for proper protection.

5.2 Marking.

5.2.1 Special marking. Marking of outer containers shall be in accordance with Code of Federal Regulations 49 CFR 100-199.

5.2.2 Normal marking. Unless otherwise specified in the contract or purchase order (see 6.2), the marking on the inner, intermediate, and outer containers shall be as specified below. The specified markings shall be applied to the containers in accordance with the applicable provisions of MIL-STD-129.

a. Inner Container Marking

(1) NSN 1377-01-390-0377-MD58
(2) Cartridge, Impulse, CCU-123/E
(3) Drawing 837AS247
(4) Quantity
(5) Lot number in accordance with MIL-STD-1168
(6) DOT Explosive Class 1.1D
(7) "LEA-20 CHARGES"

b. Outer Container Marking

(1) NSN 1377-01-390-0377-MD58
(2) Cartridge, Impulse, CCU-123/E
(3) Drawing 837AS247
(4) Quantity
(5) Gross weight and cube

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(6) Lot number in accordance with MIL-STD-1168
(7) Contract/purchase order number
(8) DOT Explosive Class 1.1D
(9) UN Number 0048
(10) Proper UN name, CHARGES, DEMOLITION
(11) Performance oriented packaging identification
(12) "LEA-20 CHARGES"

5.2.3 Additional special markings. Additional special markings of outer containers shall be in accordance with 49 CFR 100-199 and shall include two messages: the NIIN and the lot number, shelf life expiration date, and quantity, which shall be encoded in a single message. The bar code label shall be applied to the outer container.

5.2.4 Bar coding. Unless otherwise specified in the contract or purchase order (see 6.2), the outer container shall be bar coded in accordance with MIL-STD-129, Part 2. Additionally, the above specified marking information shall be applied to the containers in accordance with the applicable provisions of MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended Use. The cartridge is intended to explode and split open the end of a tube to form tines and produce an underground camouflet in the earth around the end of the tube. This cartridge is used in the LEA-20 Lightweight Earth Anchor, which is used in the construction of expeditionary airfields.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2).

c. Production test activity (see 4.1).

d. Directions for shipping radiographs and certifications of entire production lot and primary component certification and test data to the activity conducting production lot acceptance (see 4.3.2 and 4.3.3).

e. Whether first article inspection is required and, if so, specify the test activity (see 3.1 and 4.2.1).

f. Number of inner containers per pack if not optional (see 5.1.1).

g. Marking if other than as specified (see 5.2.2).

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h. If bar coding is not required on outer container (see 5.2.4).

i. That the safety requirements of the "Contractor's Safety Manual for Ammunition, Explosives and Related Dangerous Material" (DOD 4145.26M) are applicable (see 6.5).

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, standard production item from the contractor's current inventory (see 3.1) and the number of items to be tested as specified in 4.2.1. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Loaded assembly hazard information.

DOD, DOT, NATO Hazard Classification - Class 1, Division 1
United Nation Serial Number - 0048
Storage Compatibility Grouping - D
DOD Markings - Charges, Demolition

6.5 Safety precautions. The safety precaution requirements of the "Contractor's Safety Manual for Ammunition and Explosives" (DOD 4145.26M), are applicable and should be specified in the contract as required by the Federal Acquisition Regulation (FAR) 23.3. NOTE: When this specification is used as part of the description of work to be accomplished by a Government activity, the safety requirements of "Ammunition and Explosives Ashore" OP 5 are applicable.

6.6 Definitions.

6.6.1 Level A packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration, and physical damage during shipment, handling, indeterminate storage, and world-wide redistribution.

6.6.2 Level A packing. The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage, and world-wide distribution.

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6.7 Subject term (key word) listing.

Cartridge
Charge
Earth Anchor
LEA-20
MD58

Preparing Activity:
Navy - OS
(Project 1377-NE61)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-82922(OS)	941115

3. DOCUMENT TITLE
CARTRIDGE, IMPULSE, CCU-123/E (EARTH ANCHOR CARTRIDGE)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDER, INDIAN HEAD DIVISION NAVAL SURFACE WARFARE CENTER (CODE 8420)	(1) Commercial 301-743-4358/4510	(2) AUTOVON 354-4358/4510
c. ADDRESS (Include Zip Code) 101 STRAUSS AVE. INDIAN HEAD, MD 20640-5035	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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